FINAL PROJECT REPORT
BAN 620 - DATA MINING

Project Title:
" Predicting Credit Card Defaults: A Data-Driven Approach to Financial Risk Management "

Group 2

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Introduction to the Industry:
The credit card industry forms a critical pillar of modern finance, enabling individuals and businesses to access flexible credit for transactions and investments. However, the industry also faces significant financial risks due to payment defaults, which can lead to substantial losses for financial institutions. Effective credit risk management is essential to mitigate these risks, ensuring the stability and sustainability of credit systems. Predictive analytics has become an invaluable tool in identifying patterns and trends associated with defaults, helping financial institutions make informed decisions regarding credit approvals and customer risk assessments.

Taiwan's Credit Card Crisis: A Backstory
In the 1990s, the Taiwanese government allowed the establishment of new banks, leading to aggressive lending practices, especially in real estate. When the real estate market stagnated, banks turned to credit and cash cards as new revenue streams. They launched aggressive advertising campaigns, eased approval requirements, and targeted young people, including students with little or no income, encouraging widespread credit usage.
By 2006, credit card debt skyrocketed to $268 billion USD, with over half a million individuals unable to repay their loans. These "credit card slaves" could only afford the minimum monthly payment, regardless of their bill amounts, leading to a vicious debt cycle. This inability to repay caused severe societal problems, including homelessness, family breakdowns, and an increase in suicide rates.
This financial landscape contextualizes our analysis, particularly the insight that many individuals paid only the minimum amount monthly, which is evident in our data visualization. This historical backdrop highlights how unchecked lending practices contributed to unsustainable debt behaviors in Taiwan.

About the Dataset:
Our project leverages a dataset derived from an actual Taiwan bank in 2005, focusing on the factors influencing credit card defaults.

Link to the dataset: https://www.kaggle.com/datasets/narayanach/default-of-credit-card-clients

This dataset offers insights into customer demographics and financial behaviors, making it an excellent resource for exploring predictive modeling in financial risk management.
The dataset contains 25 variables, encompassing basic demographic details such as **marital status**, **education level**, and **gender**, alongside detailed financial records, including monthly bill amounts, payments made, and repayment statuses over a six-month period. These variables provide a comprehensive view of customer profiles, enabling an analysis of default patterns.

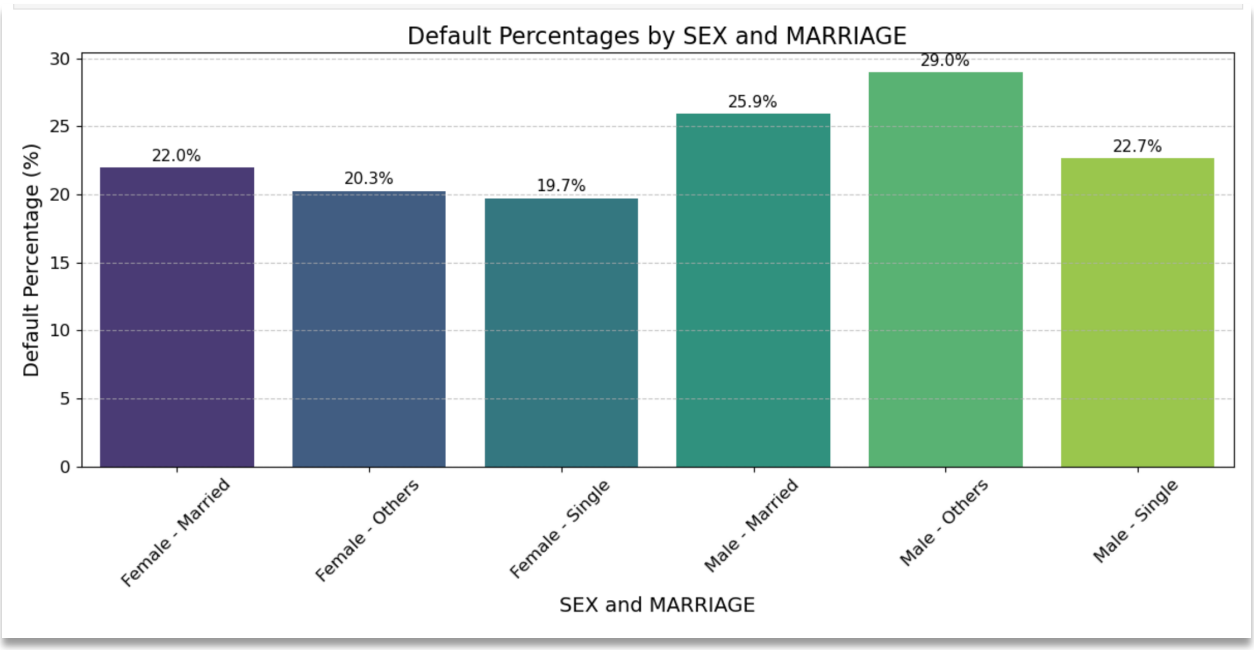
Objective:
The primary aim of this project is to use data-driven approaches to predict the likelihood of credit card defaults. By identifying the strongest predictors of defaults and examining the relationship between demographic and financial variables and default rates, we aim to enhance the understanding of risk factors within this domain.
Inspiration for Exploration:

- How do default probabilities vary across different demographic categories such as age, gender, and marital status?
- Which financial variables, such as repayment status or bill amounts, have the most significant impact on default likelihood?

<u>Data Preprocessing</u>
The dataset underwent a thorough cleaning process, ensuring no missing values and improving usability by renaming columns for clarity. Numerical values in repayment status, education, and marriage columns were mapped to descriptive categories (e.g., "-1" to "Paid full," "1" to "Graduate school," etc.), enhancing interpretability. Additionally, one-hot encoding was applied to categorical variables like gender, education, marital status, and repayment statuses to prepare the data for analysis. This streamlined process ensured a clean, well-structured dataset for predictive modeling.

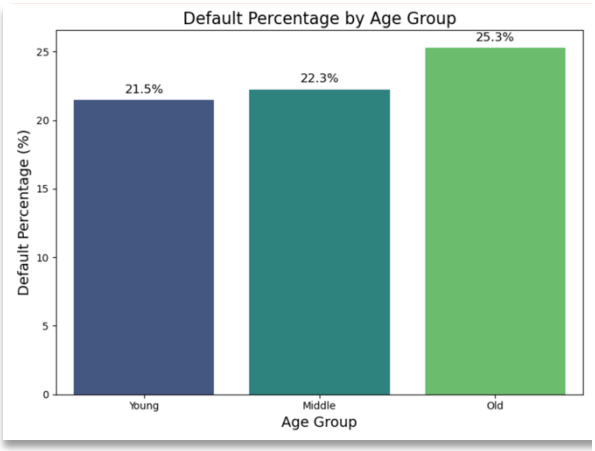
DATA VISUALIZATION:

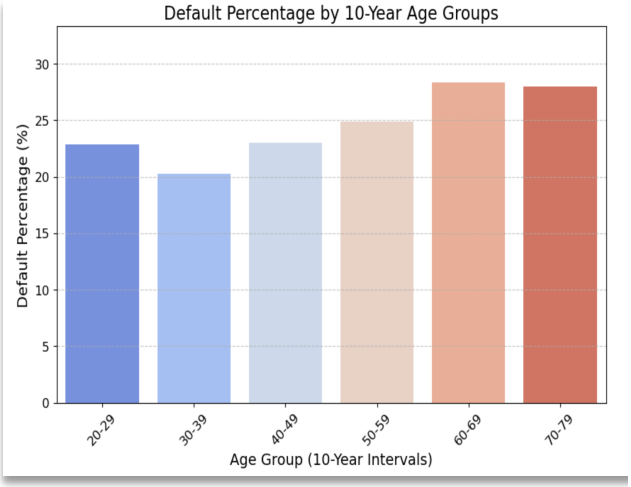
GENDER and MARITAL STATUS:



The data shows that Male - Others have the highest default rate at 29.0%, however they are very few in number in our entire dataset and also the category others is not very well defined, so we cannot infer much from it.
Therefore, the next highest default rate is from Married Male(25.9%) and the lowest default rate is from Female - Single (19.7%).

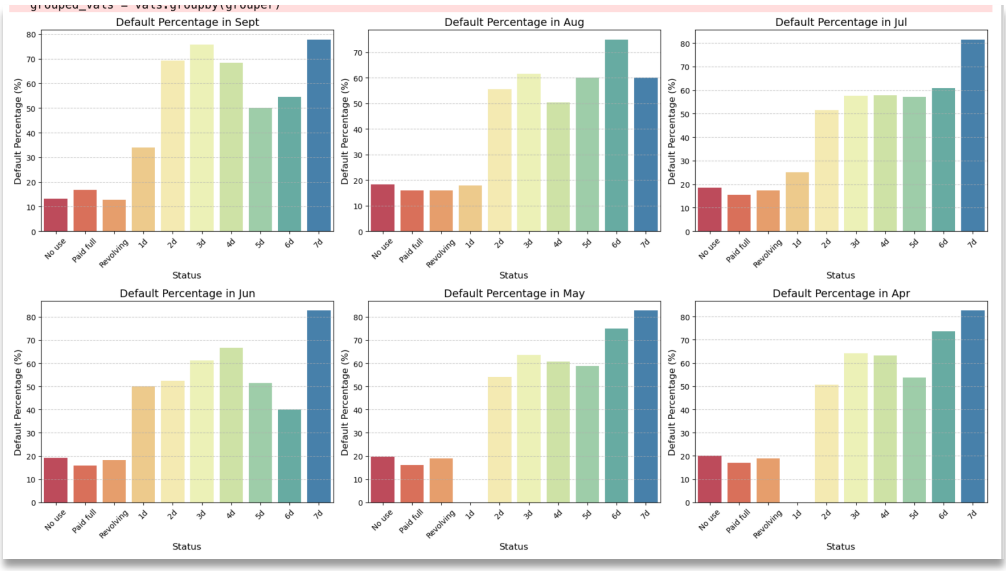
AGE GROUPS:





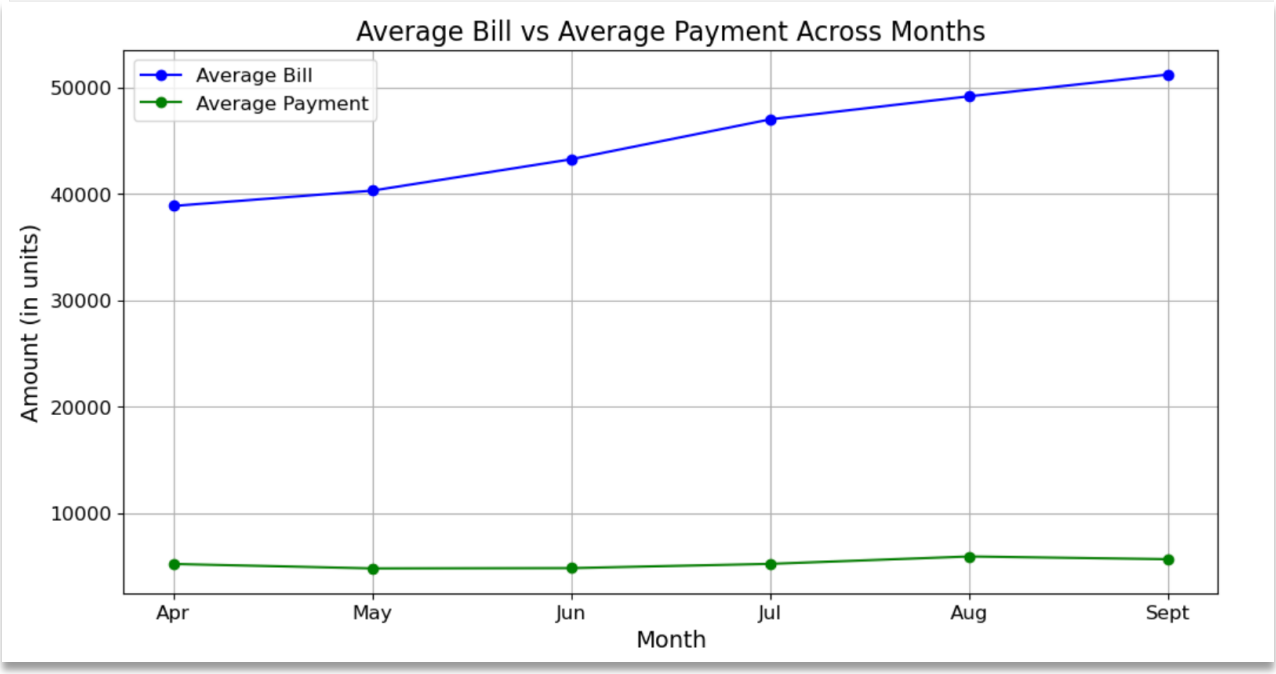
The default percentages increase with age. Younger individuals (20–29) have a lower default rate (approx. 23%), which rises gradually for middle-aged groups (50–59) and peaks at approx.. 29% for older individuals (60–79), indicating that older age groups are more prone to defaults.

ACROSS MONTHS:



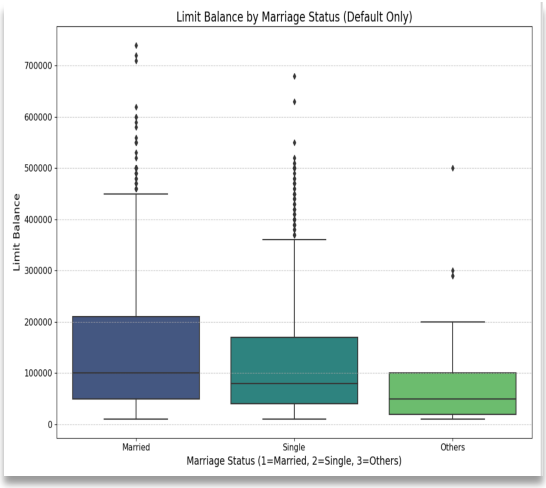
"7d" (7 days delayed) consistently has the highest default percentage showing a stable trend across months.
"No use" and "Paid full" have the lowest percentages with minimal month-to-month changes.
"1d" to "4d" (1–4 days delayed) exhibit moderate defaults, with slight monthly variations but no drastic shifts.
Overall, the trend across repayment statuses is consistent, with severe delays (e.g., "7d") showing no significant drop or rise month-to-month.

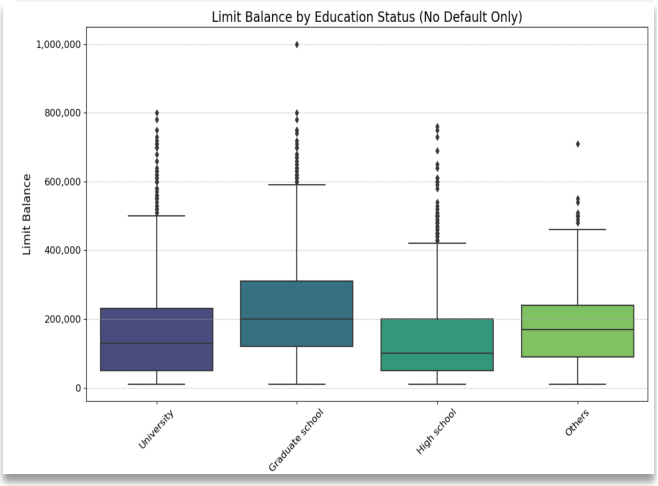
Average bill vs Average Payment:



The graph shows that average payments remain relatively constant across months despite the steady increase in average bills. This trend can be linked to the backstory of the dataset, where many individuals in Taiwan during the credit card crisis were only able to make the minimum payments on their bills, regardless of their outstanding balances. This behavior likely resulted from aggressive lending practices and targeting of individuals with limited income, creating a vicious debt cycle where payments fail to keep up with rising bill.

LIMIT BALANCE:

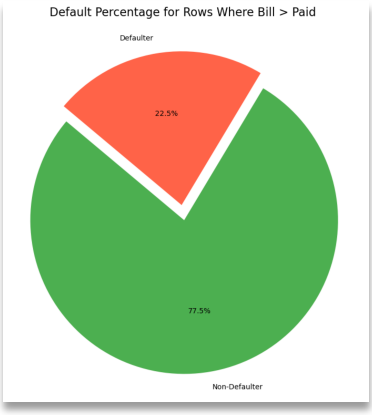




If we look at the default profile from marriage status, most default customers are married (and they were given higher limit balance too). I may assume that this is because they may have higher bill/monthly expenses than those who are singles, making them difficult in paying the loan.

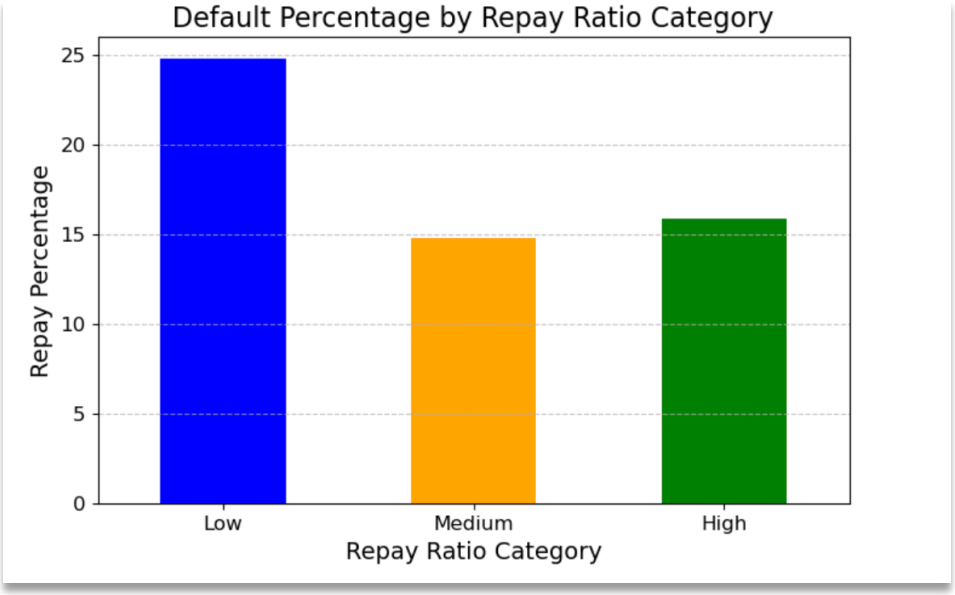
The graph shows that individuals with Graduate School and University education levels have the highest limit balances, with wider spreads and notable outliers, indicating they are given higher credit limits. This could be due to their perceived higher earning potential or financial stability. In contrast, those with High School education have the lowest median limit balances, suggesting a lower level of financial trust or capacity. This reflects how credit limits are often aligned with education levels, which are seen as indicators of repayment ability.

<u>BILL > PAID:</u>



The higher non-default rate (77.5%) despite bills exceeding payments suggests that the dataset might classify defaults based on stricter criteria, such as prolonged non-repayment or higher overdue amounts, rather than a simple bill-payment mismatch.

REPAYMENT RATIO:



From this graph, we can infer that individuals with lower repayment ratios are at a higher risk of default, likely due to limited financial capacity or reliance on minimum payments. Higher repayment ratios are associated with lower default rates, indicating better financial management and repayment behavior.

3. TRYING DIFFERENT MODELS:

Model 1: Using all possible predictors

Model 2: HANDLING
'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6' COLUMN VALUES

```
if value in [-2, -1, 0]:
    return 'no_issue'     # AS THEY WONT CAUSE A DEFAULT
elif value == 1:
    return 'first_time_default'
elif value in [2, 3]:
    return 'short_term_default'
else:
    return 'long_term_default'
```

Predictor variables : 'zLIMIT_BAL', 'zAGE', 'zbill_sept', 'zbill_aug', 'zbill_jul','zbill_jun', 'zbill_may', 'zbill_apr', 'zpaid_sept', 'zpaid_aug', 'zpaid_jul', 'zpaid_jun', 'zpaid_may', 'zpaid_apr', 'EDUCATION_High_school', 'EDUCATION_Others', 'EDUCATION_University', 'MARRIAGE_Married', 'MARRIAGE_Single', 'status_sept_first_time_default', 'status_sept_long_term_default', 'status_sept_short_term_default', 'status_aug_first_time_default',

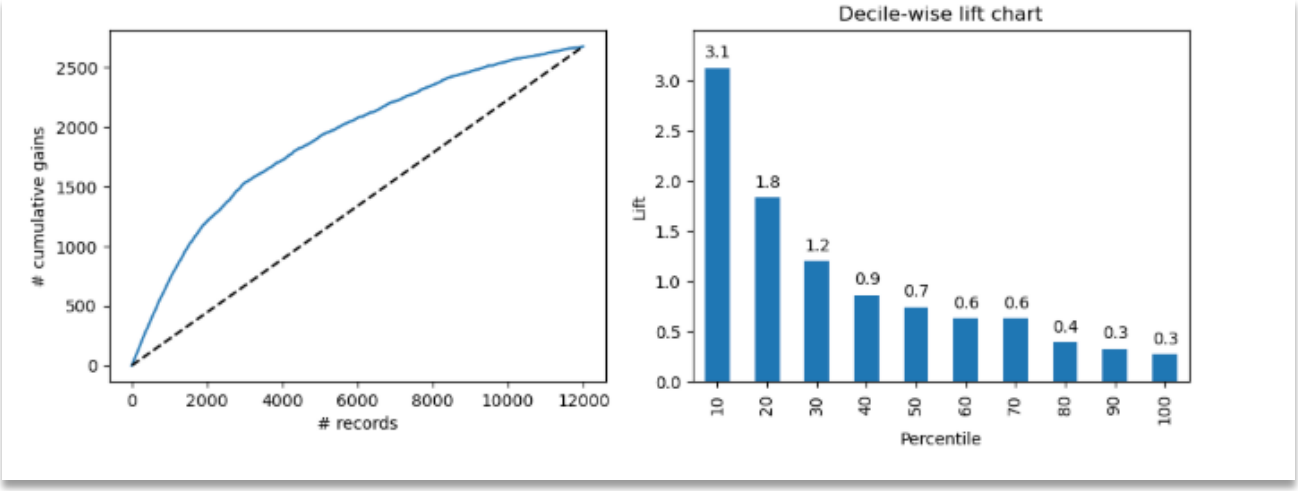
'status_aug_long_term_default','status_aug_short_term_default',
'status_jul_first_time_default', 'status_jul_long_term_default',
'status_jul_short_term_default','status_jun_first_time_default', 'status_jun_long_term_default',
'status_jun_short_term_default',
'status_may_long_term_default','status_may_short_term_default',
'status_apr_long_term_default','status_apr_short_term_default'
Target variable: 'default'

Model 3: HANDLING
'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6' COLUMN VALUES

```
 if value == -2:
      return 'overpaid'
   elif value == -1:
      return 'on_time'
   elif value == 0:
      return 'neutral'
   elif value == 1:
      return '1_month_default'
   elif value in [2,3]:
      return 'short_default'
   elif value in [4, 5]:
      return 'medium_default'
   else:  # 6, 7, 8, 9
      return 'severe_default'
```

Predictor variables : 'LIMIT_BAL', 'SEX', 'AGE', 'bill_sept', 'bill_aug', 'bill_jul', 'bill_jun', 'bill_may', 'bill_apr', 'paid_sept', 'paid_aug', 'paid_jul','paid_jun', 'paid_may', 'paid_apr', 'EDUCATION_High school','EDUCATION_Others', 'EDUCATION_University', 'MARRIAGE_Others','MARRIAGE_Single', 'status_sept_medium_default', 'status_sept_neutral','status_sept_on_time', 'status_sept_overpaid','status_sept_severe_default', 'status_sept_short_default', 'status_aug_medium_default', 'status_aug_neutral', 'status_aug_on_time','status_aug_overpaid', 'status_aug_severe_default','status_aug_short_default', 'status_jul_medium_default', 'status_jul_neutral', 'status_jul_on_time', 'status_jul_overpaid','status_jul_severe_default', 'status_jul_short_default','status_jun_medium_default', 'status_jun_neutral', 'status_jun_on_time','status_jun_overpaid', 'status_jun_severe_default','status_jun_short_default', 'status_may_neutral', 'status_may_on_time','status_may_overpaid', 'status_may_severe_default','status_may_short_default', 'status_apr_neutral', 'status_apr_on_time','status_apr_overpaid', 'status_apr_severe_default', 'status_apr_short_default'
Target variables: 'default'

Model 1 logistic regression interpretation:



From the above picture, this model shows a significant lift compared to random selection, especially in the first 10% and second 10% validation data, which is 3.1 and 1.8 times higher than random selection to capture default. Top 20% validation data identify 50% of default.

Therefore, the focus should be on the top decile for targeted interventions as they provide the maximum return on investment. Prioritize resources for high-risk customers identified in the top decile.

1. **High-Risk Customers (Top 10% Decile)**:
    - Proactively engage these customers with strategies like payment reminders, restructuring payment plans, or offering incentives for timely payments.
    - Consider adjusting their credit limits or terms to mitigate potential losses.
2. **Moderate-Risk Customers (2nd and 3rd Deciles)**:
    - Use softer interventions such as education on financial planning or offering lower-interest repayment plans to prevent them from escalating to high risk.
3. **Low-Risk Customers (Bottom Deciles)**:
    - Focus on retention and promotional offers, such as increased credit limits or lower fees, to enhance loyalty without significant risk.

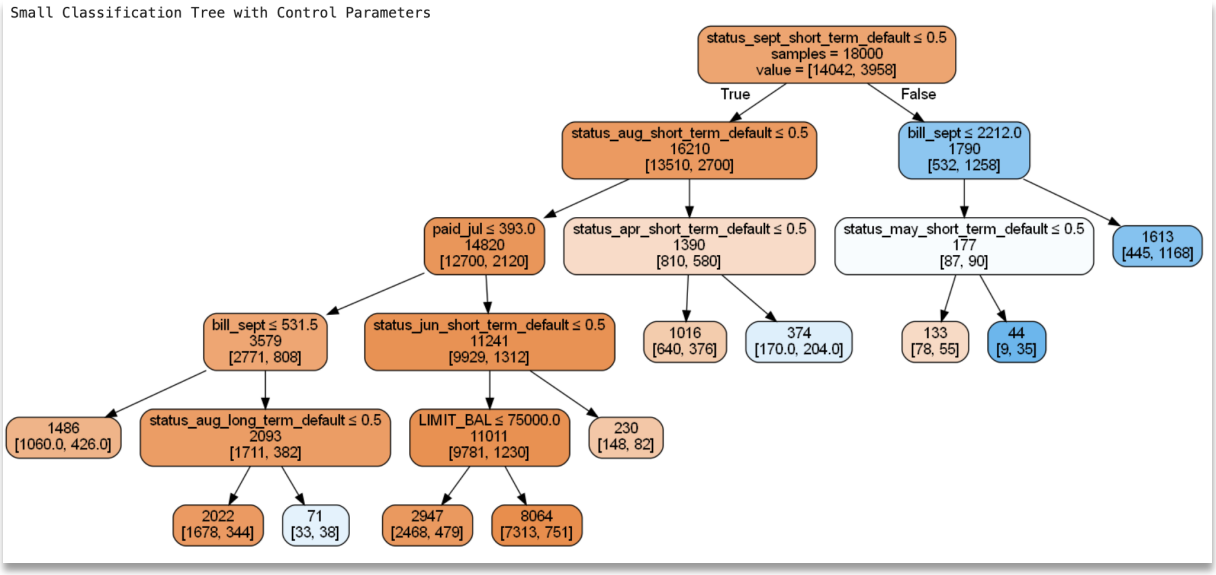
```
Classification Summary by Cutoff
   Cutoff  Accuracy  Precision    Recall  Specificity   ROC-AUC    TP    FP  \
0     0.1  0.371750   0.256194  0.953697     0.204570  0.768791  2554  7415
1     0.2  0.761417   0.472850  0.601568     0.807337  0.768791  1611  1796
2     0.3  0.802583   0.568212  0.480583     0.895087  0.768791  1287   978
3     0.4  0.817333   0.647451  0.398432     0.937674  0.768791  1067   581
4     0.5  0.818417   0.677833  0.355116     0.951513  0.768791   951   452
5     0.6  0.815083   0.708069  0.291636     0.965458  0.768791   781   322
6     0.7  0.802417   0.743265  0.175131     0.982622  0.768791   469   162
7     0.8  0.793167   0.775281  0.103062     0.991418  0.768791   276    80
8     0.9  0.777667   0.708333  0.006348     0.999249  0.768791    17     7

     TN    FN
0  1907   124
1  7526  1067
2  8344  1391
3  8741  1611
4  8870  1727
5  9000  1897
6  9160  2209
7  9242  2402
8  9315  2661
```

ROC-AUC remains consistent across cutoffs (**0.7688**), indicating overall model performance is stable in distinguishing defaults from non-defaults.
At lower cutoffs (e.g., 0.1), the model captures almost all defaults (high recall), but loses many false positives (low precision). At higher cutoffs (e.g., 0.9), precise in predicting defaults (high precision) but misses many actual defaults (low recall), among them a cutoff of **0.5 has highest overall accuracy of 0.8184, and it also balances accuracy, precision, and recall, making it suitable for operational decisions where both default and non-default classifications are important but it depends on the business goals to choose proper cutoff.**

1. **For Proactive Interventions**:
    - Use a lower cutoff (e.g., 0.2–0.3) to identify as many high-risk customers as possible.
    - Offer payment plans, reminders, or credit adjustments to reduce default risk.
2. **For Cost-Effective Risk Mitigation**:
    - Use a moderate cutoff (e.g., 0.5) to balance between catching defaults and minimizing false positives.
    - Target the identified customers with tailored financial counseling.
3. **For Marketing or Retention Campaigns**:
    - Use a higher cutoff (e.g., 0.7–0.8) to focus on customers most likely to repay, reducing marketing costs and improving campaign efficiency.

DECISION TREE:



From the decision tree, it is evident that as soon as there starts a short term delay i.e., 2- or 3-month delays in payments, the likelihood of defaulting in the next month significantly increases (moving towards the right branch of the graph). Conversely, if the delay is less severe, such as a first-time default, the chances of defaulting are considerably lower.

4. MODEL SELECTION:

Based on the analysis of the confusion matrices for various cutoff thresholds, **Logit Model 3** with a cutoff of **0.3** emerges as the most suitable choice for this case study. This decision is informed by the context of the **Taiwanese credit card dataset**, where the primary goal is to effectively predict defaults, given their critical impact on the organization's financial health

1. Sensitivity is Critical for Predicting Defaults

In the domain of credit card lending, accurately predicting **defaults** (true positives) is important because:

- A **missed default** (false negative) represents a significant financial risk to the bank. Customers who default can lead to unrecovered debts, damaging the organization's profitability.
- On the other hand, a **false positive** (non-default predicted as default) might lead to some operational inefficiencies (e.g., denying credit to a customer who would repay), but the financial risk is comparatively lower.

Logit Model 3 achieves the highest sensitivity (48.2%), making it the best choice for minimizing false negatives and ensuring that most potential defaulters are identified.

2. Trade-off Between Sensitivity and Specificity

While **specificity** decreases to (89.4%), this trade-off is acceptable because:

- The primary business objective is to mitigate financial losses from defaults.
- False positives (non-defaulters flagged as defaulters) may lead to lost revenue opportunities, but they are far less harmful than defaults that are not predicted.

In this case, prioritizing sensitivity aligns better with the organization's risk management strategy, even if it means a slight reduction in specificity.

3. In context to the Taiwan dataset highlights a unique challenge in credit card lending:

- Banks issued credit cards with lenient approval criteria, targeting young customers who often lacked sufficient income.
- This resulted in a higher likelihood of defaults, making the early detection of default risks even more critical.

Conclusion
Logit Model 3, with its higher sensitivity, is the optimal choice for this case study. While it sacrifices some specificity, this trade-off is justified because accurately predicting defaults has a more significant impact on mitigating financial risks. By prioritizing sensitivity, the model supports the organization's goals of reducing bad debts while enabling proactive, ethical risk management in a challenging market environment.

POTENTIAL IDEA TO STREAMLINE THE RISK MITIGATION:

Summary of CHI Model A:

Objective

The **Credit History Index (CHI)** Model A is designed to assess credit risk by categorizing clients into risk groups based on their **repayment behavior**, **credit utilization**, and **payment-to-bill ratios**. It successfully identifies clients at high, moderate, and low risk of default.

CHI Calculation Logic

1. **Starting Score**: 70 points.
2. **Adjustments**:
    - **Payment Behavior (PAY_0 to PAY_6)**:
        - Timely Payment (-1): **+3 points**.
        - 1-Month Delay: **-6 points**.
        - 2–3 Months Delay: **-12 points**.
        - 4+ Months Delay: **-24 points**.
    - **Credit Utilization**:

        - 0–30% Utilization: **+0 points** (neutral).
        - 30–70% Utilization: **+5 points** (reward moderate usage).
            - 70% Utilization: **-8 points** (penalty for high usage).
    - **Payment-to-Bill Ratio**:
        - ≥90% Payment Ratio: **+6 points**.
        - 50–90% Payment Ratio: **+3 points**.
        - <50% Payment Ratio: **-6 points**.
3. **Score Range**: 0–120 points.

CHI Tag Categories

Based on the CHI score, clients are classified into three categories:

1. **High-Risk**: CHI < 20 (Highest default rates).
2. **Moderate-Risk**: CHI 20–60 (Moderate default rates).
3. **Low-Risk**: CHI > 60 (Lowest default rates).

Performance of CHI Model A

- **High-Risk Clients**: ~65% default rate (immediate intervention needed).
- **Moderate-Risk Clients**: ~28% default rate (monitor and guide).
- **Low-Risk Clients**: ~12% default rate (reliable and profitable segment).

Customer Distribution

- **Low-Risk Clients**: 53.6%
- **Moderate-Risk Clients**: 39.1%
- **High-Risk Clients**: 7.3%

Strengths of CHI Model A

1. Effectively separates clients by default risk.
2. Encourages responsible usage and repayment behavior.
3. Provides actionable insights for intervention and reward strategies.

Conclusion

The CHI Model A is a robust and reliable tool for predicting default risk. It enables credit card companies to:

- **Target High-Risk clients** for proactive interventions.
- **Guide Moderate-Risk clients** to improve their financial habits.
- **Reward Low-Risk clients** to strengthen loyalty and profitability.

Steps and Actions for Each CHI Tag

The CHI Model A categorizes clients into **High-Risk**, **Moderate-Risk**, and **Low-Risk** based on their default probability. Below are tailored steps and actions for each group to minimize risks and maximize engagement.

1. High-Risk Clients (CHI < 20)

Objective: **Minimize defaults** and control financial exposure.

Action	Details
Early Intervention	- Send automated reminders (calls, emails, SMS) before payment due dates.
Debt Counseling	- Offer financial literacy sessions and personalized debt repayment advice.
Temporary Credit Limit Reduction	- Reduce credit limits to prevent further debt accumulation.
Structured Repayment Plans	- Offer flexible repayment plans (e.g., partial payments, EMIs) to help clear overdue balances.
Penalty Waiver Programs	- Waive penalties (e.g., late fees) if clients pay a significant portion of overdue amounts.
Close Monitoring	- Flag accounts for frequent monitoring to detect further deterioration in repayment behavior.

:

2. Moderate-Risk Clients (CHI 20–60)

Objective: **Guide behavior improvement** and prevent clients from slipping into high-risk.

Action	Details
Incentivize CHI Improvement	- Offer rewards (e.g., cashback, reduced interest rates) for consistent on-time payments.
Interest Rate Discounts	- Reduce interest rates for clients paying ≥70% of their bill regularly over 3-6 months.
Educational Insights	- Send monthly reports explaining their CHI, what affects it, and actionable tips for improvement.
Flexible Payment Options	- Allow installment plans for larger bills to avoid overwhelming clients financially.
Spend-Based Rewards	- Reward responsible usage (e.g., controlled spending with timely payments).

Action	Details
Credit Monitoring Alerts	- Implement real-time alerts for rising credit utilization or nearing due dates.

3. Low-Risk Clients (CHI > 60):

Objective: **Retain and reward** financially disciplined clients to maximize engagement and profitability.

Action	Details
Credit Limit Increases	- Gradually increase credit limits to encourage higher card usage while maintaining discipline.
Loyalty Rewards Programs	- Offer exclusive benefits like cashback, bonus points, and merchant discounts for active usage.
Promote Premium Products	- Upsell premium credit cards with better perks, such as travel rewards, dining offers, etc.
Interest-Free Period Offers	- Introduce short-term interest-free periods for large purchases to increase spending engagement.
Spending Bonuses	- Provide spending-based bonuses (e.g., cashback for reaching spending milestones).
Exclusive Client Services	- Offer priority customer support, waived annual fees, and exclusive merchant partnerships.

Summary of Actions by CHI Tag:

CHI Tag	Objective	Key Actions
High-Risk	Minimize Defaults	Early intervention, credit limit reduction, structured repayment plans, and close monitoring.
Moderate-Risk	Guide Behavior	Incentives for CHI improvement, flexible payments, interest rate reductions, and education.
Low-Risk	Retain and Reward	Credit limit increases, loyalty rewards, premium product offers, and spend-based bonuses.

Conclusion:

By taking **targeted actions** for each CHI category, credit card companies can:

- **Reduce defaults** among High-Risk clients.
- **Improve repayment behavior** for Moderate-Risk clients.
- **Maximize engagement and profitability** from Low-Risk clients.

This strategy ensures a balanced approach to risk management and customer satisfaction.

Additional recommendations:

- Enhanced Application Screening: Strengthen approval processes with rigorous income verification, job stability criteria, and multi-tier income thresholds for credit limits.
- Early Intervention Programs: Send automated reminders (calls, emails, SMS) before payment due dates to reduce missed payments.
- Flexible Repayment Solutions: Offer structured repayment plans and penalty waiver programs to help clients manage overdue balances.
- Proactive Credit Limit Management: Temporarily reduce credit limits for high-risk clients and flag accounts for close monitoring to detect repayment issues early.

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